Space Vector Modulation for a Two-Level Three-Phase Inverter

From (U_α, U_β) to Gate Signals:
Classical SVM and Triangular Third-Harmonic Injection

Abstract

This document describes how to generate the switching duty cycles of a two-level three-phase voltage source inverter starting from a voltage reference expressed in the stationary $\alpha\beta$ frame. Two implementations are presented: the classical sector-based space vector modulation (SVM) and the carrier-based scheme with triangular (min–max) third-harmonic injection. The two are shown to be mathematically identical when the zero vector is split symmetrically, which makes the second form the preferred one for embedded implementation.

Contents

1 Inverter model and notation **2**
- 1.1 Switching states . . . 2
- 1.2 Clarke transformation . . . 2
- 1.3 The eight voltage vectors . . . 2

2 Reference vector and linear modulation range **2**
- 2.1 Modulation index . . . 3

3 Classical space vector modulation **3**
- 3.1 Principle . . . 3
- 3.2 Sector identification . . . 3
- 3.3 On-times: closed form . . . 4
- 3.4 Auxiliary-variable formulation . . . 5
- 3.5 Zero-vector split and switching sequence . . . 5
- 3.6 From on-times to duty cycles . . . 5

4 Triangular third-harmonic injection (min–max) **6**
- 4.1 Zero-sequence degree of freedom . . . 6
- 4.2 The min–max (triangular) choice . . . 6
- 4.3 Properties of the injected waveform . . . 6
- 4.4 Why the range extends . . . 7
- 4.5 Comparison with sinusoidal third-harmonic injection . . . 7

5 Equivalence of the two methods **8**

6 Implementation **8**
- 6.1 Classical SVM . . . 8
- 6.2 Min–max injection . . . 9

7 Practical notes **9**

8 Summary **10**

1 Inverter model and notation

1.1 Switching states

The two-level three-phase bridge has three legs, each described by a binary switching function $s_x \in \{0,1\}$, $x \in \{a,b,c\}$, where $s_x = 1$ means the upper device of leg x is conducting. The pole voltage referred to the negative DC rail is

$$u_{x0} = s_x V_{\mathrm{dc}}. \tag{1}$$

For a balanced, isolated-neutral three-phase load, the phase voltage referred to the load neutral N is

$$u_{xN} = V_{\mathrm{dc}} \left(s_x - \frac{s_a + s_b + s_c}{3} \right). \tag{2}$$

1.2 Clarke transformation

The amplitude-invariant Clarke transformation is used throughout:

$$\begin{bmatrix} U_\alpha \\ U_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix}, \qquad \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} U_\alpha \\ U_\beta \end{bmatrix}. \tag{3}$$

With this scaling a balanced set of phase voltages of peak value $\hat{U}$ maps to a space vector of constant magnitude $\hat{U}$ rotating at ω.

1.3 The eight voltage vectors

Applying (2) and (3) to the eight switching combinations yields six *active* vectors of equal magnitude $\frac{2}{3}V_{\mathrm{dc}}$, spaced 60° apart, and two *zero* vectors:

$$\boldsymbol{v}_k = \frac{2}{3} V_{\mathrm{dc}}\, e^{j(k-1)\pi/3}, \qquad k = 1 \ldots 6, \qquad \boldsymbol{v}_0 = \boldsymbol{v}_7 = 0. \tag{4}$$

Vector	$s_a s_b s_c$	u_{aN}	u_{bN}	u_{cN}	Complex value
$\boldsymbol{v}_0$	000	0	0	0	0
$\boldsymbol{v}_1$	100	$\frac{2}{3}V_{\mathrm{dc}}$	$-\frac{1}{3}V_{\mathrm{dc}}$	$-\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j0}$
$\boldsymbol{v}_2$	110	$\frac{1}{3}V_{\mathrm{dc}}$	$\frac{1}{3}V_{\mathrm{dc}}$	$-\frac{2}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j\pi/3}$
$\boldsymbol{v}_3$	010	$-\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}$	$-\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j2\pi/3}$
$\boldsymbol{v}_4$	011	$-\frac{2}{3}V_{\mathrm{dc}}$	$\frac{1}{3}V_{\mathrm{dc}}$	$\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j\pi}$
$\boldsymbol{v}_5$	001	$-\frac{1}{3}V_{\mathrm{dc}}$	$-\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j4\pi/3}$
$\boldsymbol{v}_6$	101	$\frac{1}{3}V_{\mathrm{dc}}$	$-\frac{2}{3}V_{\mathrm{dc}}$	$\frac{1}{3}V_{\mathrm{dc}}$	$\frac{2}{3}V_{\mathrm{dc}}\, e^{j5\pi/3}$
$\boldsymbol{v}_7$	111	0	0	0	0

Table 1: Switching states of the two-level bridge.

2 Reference vector and linear modulation range

The controller output is the reference vector

$$\boldsymbol{u}^* = U_\alpha + j\, U_\beta = |\boldsymbol{u}^*|\, e^{j\theta}, \qquad |\boldsymbol{u}^*| = \sqrt{U_\alpha^2 + U_\beta^2}, \qquad \theta = \operatorname{atan2}(U_\beta, U_\alpha). \tag{5}$$

The six active vectors define a hexagon. Since the local average of the inverter output over one switching period is a convex combination of the applied vectors, any $\boldsymbol{u}^*$ inside the hexagon is realisable. To keep the magnitude independent of the angle (i.e. to avoid distortion), the reference must remain inside the inscribed circle:

$$|\boldsymbol{u}^*| \le \frac{2}{3} V_{\mathrm{dc}} \cos\frac{\pi}{6} = \frac{V_{\mathrm{dc}}}{\sqrt{3}} \approx 0.577\, V_{\mathrm{dc}}. \tag{6}$$

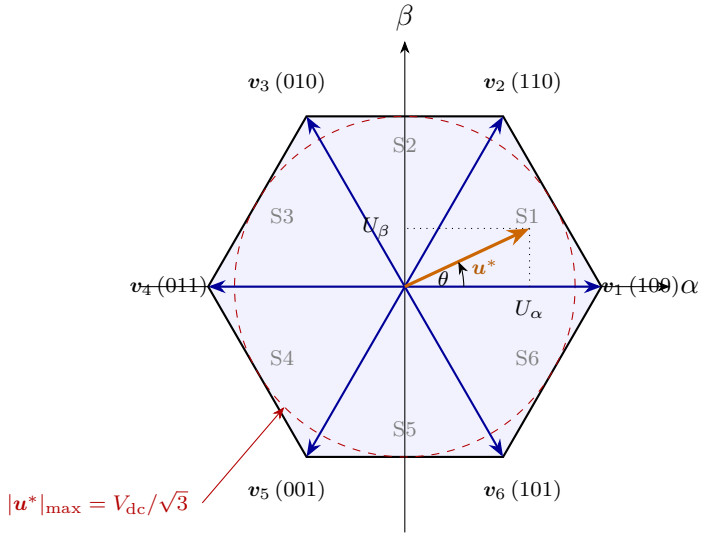


Figure 1: Hexagon of the two-level inverter, sector numbering and the inscribed circle bounding the linear modulation range. Active vector length is $\frac{2}{3}V_{\mathrm{dc}}$.

2.1 Modulation index

Two definitions are common. Referred to the linear limit:

$$m = \frac{|\boldsymbol{u}^*|}{V_{\mathrm{dc}}/\sqrt{3}} = \frac{\sqrt{3}\,|\boldsymbol{u}^*|}{V_{\mathrm{dc}}} \in [0, 1]. \tag{7}$$

Referred to the half DC link (the classical PWM definition):

$$M = \frac{\hat{U}_{\mathrm{ph}}}{V_{\mathrm{dc}}/2}, \qquad M_{\max} = \frac{2}{\sqrt{3}} \approx 1.1547. \tag{8}$$

Sinusoidal PWM without injection is limited to $\hat{U}_{\mathrm{ph}} = V_{\mathrm{dc}}/2$, so both SVM and third-harmonic injection provide a 15.47 % increase of the available fundamental voltage for the same DC link.

3 Classical space vector modulation

3.1 Principle

Over one switching period T_s the reference is synthesised as the volt-second average of the two active vectors adjacent to it, $\boldsymbol{v}_k$ and $\boldsymbol{v}_{k+1}$, plus the zero vectors:

$$\boldsymbol{u}^*\, T_s = \boldsymbol{v}_k\, t_k + \boldsymbol{v}_{k+1}\, t_{k+1}, \qquad t_0 = T_s - t_k - t_{k+1} \ge 0. \tag{9}$$

3.2 Sector identification

Two equivalent methods are used.

Angle method. Compute $\theta = \operatorname{atan2}(U_\beta, U_\alpha)$, wrap to $[0, 2\pi)$, then $k = \lfloor \theta/(\pi/3) \rfloor + 1$. Simple but requires an arctangent.

Sign method (preferred in firmware). Build the three projections of the reference on the phase axes, from the inverse Clarke transformation (3):

$$u_a = U_\alpha, \qquad u_b = -\frac{1}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta, \qquad u_c = -\frac{1}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta. \tag{10}$$

The sector boundaries at multiples of 60° coincide exactly with the sign changes of the *differences* $u_a - u_b$, $u_b - u_c$, $u_c - u_a$, since

$$u_b - u_c = \sqrt{3}\,U_\beta, \qquad u_a - u_b = \frac{3}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta, \qquad u_c - u_a = -\frac{3}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta, \tag{11}$$

which vanish respectively at $\theta = 0°, 180°$; $60°, 240°$; $120°, 300°$. Equivalently, *the sector is determined by the ordering of* u_a, u_b, u_c. Forming the binary word

$$N = 1 \cdot [u_a > u_b] + 2 \cdot [u_b > u_c] + 4 \cdot [u_c > u_a] \tag{12}$$

gives the look-up of Table 2: three additions and three comparisons, no trigonometry.

Ordering	$u_a > u_b > u_c$	$u_b > u_a > u_c$	$u_b > u_c > u_a$	$u_c > u_b > u_a$	$u_c > u_a > u_b$	$u_a > u_c > u_b$
N	3	2	6	4	5	1
Sector k	1	2	3	4	5	6

Table 2: Sector look-up from the ordering of the phase projections. Note that the same max/min information is what the min–max scheme of Section 4 uses directly.

3.3 On-times: closed form

Projecting (9) on the real and imaginary axes and solving the 2×2 system gives, for a generic sector k,

$$t_k = \frac{\sqrt{3}\,T_s}{V_{\mathrm{dc}}}\left[U_\alpha \sin\!\left(\frac{k\pi}{3}\right) - U_\beta \cos\!\left(\frac{k\pi}{3}\right)\right], \tag{13}$$

$$t_{k+1} = \frac{\sqrt{3}\,T_s}{V_{\mathrm{dc}}}\left[-U_\alpha \sin\!\left(\frac{(k-1)\pi}{3}\right) + U_\beta \cos\!\left(\frac{(k-1)\pi}{3}\right)\right], \tag{14}$$

$$t_0 = T_s - t_k - t_{k+1}. \tag{15}$$

Example: sector 1. With $k = 1$, $\boldsymbol{v}_1 = \frac{2}{3}V_{\mathrm{dc}}$ and $\boldsymbol{v}_2 = \frac{2}{3}V_{\mathrm{dc}}e^{j\pi/3}$, (9) reads

$$U_\alpha T_s = \frac{2}{3}V_{\mathrm{dc}}\left(t_1 + \frac{1}{2}t_2\right), \tag{16}$$

$$U_\beta T_s = \frac{2}{3}V_{\mathrm{dc}}\left(\frac{\sqrt{3}}{2}t_2\right), \tag{17}$$

hence

$$t_1 = \frac{T_s}{V_{\mathrm{dc}}}\left(\frac{3}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta\right), \qquad t_2 = \frac{\sqrt{3}\,T_s}{V_{\mathrm{dc}}}\,U_\beta, \tag{18}$$

in agreement with (13)–(14).

3.4 Auxiliary-variable formulation

For fixed-point implementations it is convenient to precompute three quantities independent of the sector:

$$X = \frac{\sqrt{3}\,T_s}{V_{\mathrm{dc}}}\,U_\beta, \qquad Y = \frac{T_s}{V_{\mathrm{dc}}}\left(\frac{3}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta\right), \qquad Z = \frac{T_s}{V_{\mathrm{dc}}}\left(-\frac{3}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta\right), \tag{19}$$

from which the on-times follow by a simple table (Table 3).

Sector k	1	2	3	4	5	6
t_k	$-Z$	Y	X	Z	$-Y$	$-X$
t_{k+1}	X	Z	$-Y$	$-X$	$-Z$	Y

Table 3: On-times of the two adjacent active vectors from X, Y, Z.

3.5 Zero-vector split and switching sequence

The zero interval t_0 is distributed between $\boldsymbol{v}_0$ (000) and $\boldsymbol{v}_7$ (111) according to a factor $k_0 \in [0, 1]$:

$$t_{000} = k_0\,t_0, \qquad t_{111} = (1 - k_0)\,t_0. \tag{20}$$

The choice $k_0 = 0.5$ gives *symmetric* (centre-aligned) SVM, which minimises the current ripple and produces the well-known double-edge pattern. Other choices generate the discontinuous PWM family: $k_0 = 1$ or $k_0 = 0$ give DPWMMIN and DPWMMAX, alternating values give DPWM0–DPWM3, each clamping one phase for 120° and reducing switching losses by up to one third.

For sector 1 the symmetric sequence over one period is

$$000 \to 100 \to 110 \to 111 \to 110 \to 100 \to 000, \tag{21}$$

i.e. only one leg commutates at each transition.

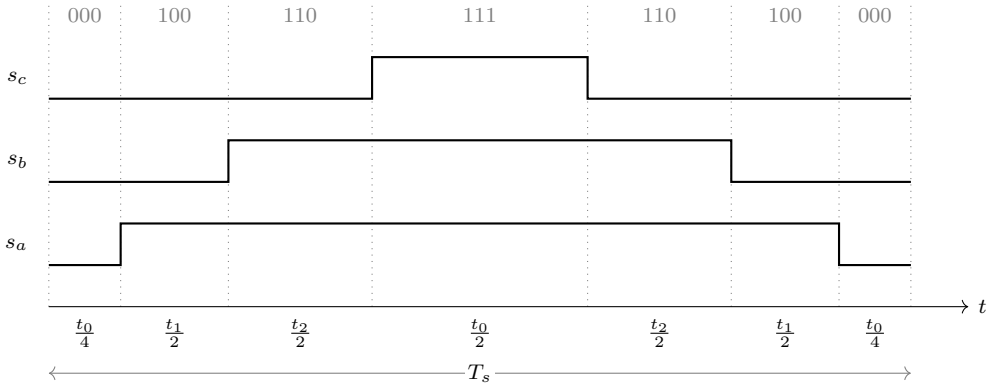


Figure 2: Symmetric (centre-aligned) switching pattern in sector 1 over one period. Only one leg commutates at each state transition; the zero time is split equally between $\boldsymbol{v}_0$ at the edges and $\boldsymbol{v}_7$ in the middle.

3.6 From on-times to duty cycles

Counting, for each leg, the total time during which its upper device is on gives the duty cycles collected in Table 4. All entries are divided by T_s.

The duty cycles are then written to the compare registers of a centre-aligned (up–down counting) timer, so that the resulting pattern is automatically symmetric about the middle of the period.

Sector	d_a	d_b	d_c
1	$(t_1 + t_2 + \frac{t_0}{2})/T_s$	$(t_2 + \frac{t_0}{2})/T_s$	$\frac{t_0}{2T_s}$
2	$(t_2 + \frac{t_0}{2})/T_s$	$(t_2 + t_3 + \frac{t_0}{2})/T_s$	$\frac{t_0}{2T_s}$
3	$\frac{t_0}{2T_s}$	$(t_3 + t_4 + \frac{t_0}{2})/T_s$	$(t_4 + \frac{t_0}{2})/T_s$
4	$\frac{t_0}{2T_s}$	$(t_4 + \frac{t_0}{2})/T_s$	$(t_4 + t_5 + \frac{t_0}{2})/T_s$
5	$(t_6 + \frac{t_0}{2})/T_s$	$\frac{t_0}{2T_s}$	$(t_5 + t_6 + \frac{t_0}{2})/T_s$
6	$(t_6 + t_1 + \frac{t_0}{2})/T_s$	$\frac{t_0}{2T_s}$	$(t_6 + \frac{t_0}{2})/T_s$

Table 4: Duty cycles of the three legs for symmetric SVM ($k_0 = 0.5$). t_k and t_{k+1} are the on-times of the two active vectors of the sector.

4 Triangular third-harmonic injection (min–max)

4.1 Zero-sequence degree of freedom

For an isolated-neutral load, the line-to-line voltages – and therefore the currents – depend only on the *differences* between the leg references. Adding the same signal u_0 to all three phase references leaves the load voltages unchanged:

$$u'_x = u_x + u_0, \qquad x \in \{a, b, c\} \quad \Longrightarrow \quad u'_{xy} = u_{xy}. \tag{22}$$

This zero-sequence component is a free design variable, and it can be used to keep the modulating signals within the linear range for a larger fundamental amplitude.

4.2 The min–max (triangular) choice

Starting from (U_α, U_β), compute the phase references with the inverse Clarke transformation:

$$u_a = U_\alpha, \qquad u_b = -\frac{1}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta, \qquad u_c = -\frac{1}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta. \tag{23}$$

Then choose the zero-sequence that centres the three references about zero:

$$\boxed{u_0 = -\frac{\max(u_a, u_b, u_c) + \min(u_a, u_b, u_c)}{2}} \tag{24}$$

and obtain the duty cycles directly:

$$\boxed{d_x = \frac{1}{2} + \frac{u_x + u_0}{V_{\mathrm{dc}}}, \qquad x \in \{a, b, c\}.} \tag{25}$$

No sector identification, no trigonometry, no look-up table: three additions, one sorting network of three elements, and three multiply-adds.

4.3 Properties of the injected waveform

For a balanced sinusoidal reference of peak $\hat{U}$ and angle $\theta = \omega t$, (24) produces a *triangular* waveform of period 120°, i.e. at three times the fundamental frequency, with peak value

$$\hat{u}_0 = \frac{\hat{U}}{4}. \tag{26}$$

Sampling a few points illustrates the shape: at $\theta = 0°$, $u_0 = -\hat{U}/4$; at $\theta = 30°$, $u_0 = 0$; at $\theta = 60°$, $u_0 = +\hat{U}/4$.

Its Fourier expansion contains only odd triplen harmonics:

$$u_0(\theta) = -\frac{2\hat{U}}{\pi^2} \sum_{n=1,3,5,\ldots} \frac{\cos(3n\theta)}{n^2} = -\,0.2026\,\hat{U}\cos 3\theta - 0.0225\,\hat{U}\cos 9\theta - \ldots \tag{27}$$

All of these are zero-sequence components: with an isolated neutral they cannot drive current and do not appear in the load voltages.

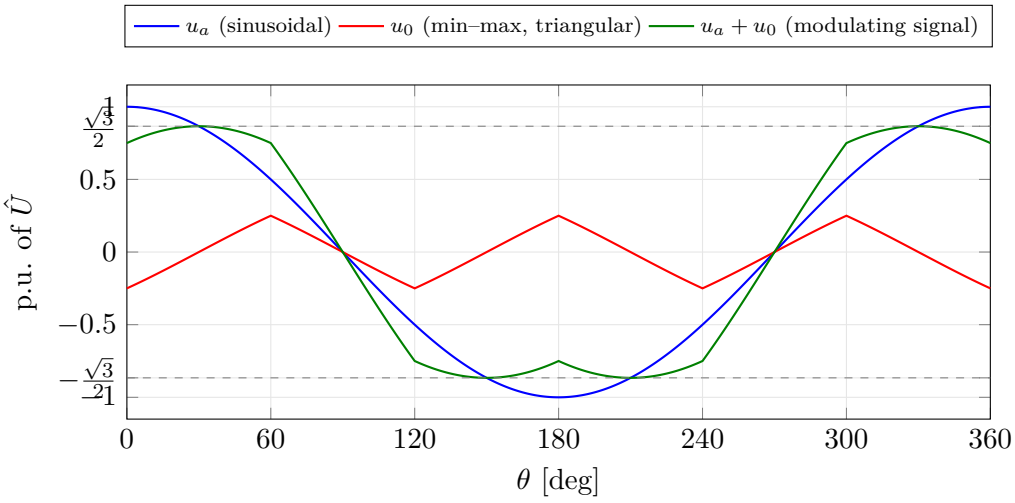


Figure 3: Sinusoidal reference of unit peak, triangular zero-sequence produced by the min–max rule, and the resulting modulating signal. The peak of the modulating signal is $\sqrt{3}/2 \approx 0.866$ instead of 1, which is exactly the 15.47 % headroom gained.

4.4 Why the range extends

The modulating signal $u_a + u_0$ has a peak of $\frac{\sqrt{3}}{2}\hat{U}$, reached at $\theta = 30°$ (Fig. 3). Linear operation requires $|u_x + u_0| \le V_{\mathrm{dc}}/2$, hence

$$\frac{\sqrt{3}}{2}\hat{U} \le \frac{V_{\mathrm{dc}}}{2} \quad \Longrightarrow \quad \hat{U} \le \frac{V_{\mathrm{dc}}}{\sqrt{3}}, \tag{28}$$

which is exactly the limit (6) obtained from the hexagon geometry.

4.5 Comparison with sinusoidal third-harmonic injection

A related scheme injects a pure sinusoid,

$$u_0 = -\frac{\hat{U}}{6}\cos 3\theta, \tag{29}$$

whose 1/6 coefficient minimises the peak of the resulting modulating signal. This also reaches $\frac{\sqrt{3}}{2}\hat{U}$ and therefore the same voltage limit, but it requires knowledge of the reference angle and amplitude and it is *not* exactly equivalent to SVM: the instantaneous ripple and the switching pattern differ slightly. The triangular (min–max) form is preferred because it needs no angle information, works instantaneously on any reference (including transients and unbalanced or non-sinusoidal references) and coincides exactly with symmetric SVM.

5 Equivalence of the two methods

Symmetric SVM with $k_0 = 0.5$ and carrier-based modulation with min–max injection produce identical duty cycles.

Proof for sector 1. In sector 1 ($0 \le \theta < 60°$) one has $u_a > u_b > u_c$, so max $= u_a$, min $= u_c$ and

$$u_0 = -\frac{u_a + u_c}{2} = -\frac{1}{2}\left(U_\alpha - \frac{1}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta\right) = -\frac{U_\alpha}{4} + \frac{\sqrt{3}}{4}U_\beta. \tag{30}$$

Equation (25) then gives

$$d_a = \frac{1}{2} + \frac{1}{V_{\mathrm{dc}}}\left(\frac{3}{4}U_\alpha + \frac{\sqrt{3}}{4}U_\beta\right). \tag{31}$$

From the SVM side, with $t_1/T_s = \left(\frac{3}{2}U_\alpha - \frac{\sqrt{3}}{2}U_\beta\right)/V_{\mathrm{dc}}$, $t_2/T_s = \sqrt{3}\,U_\beta/V_{\mathrm{dc}}$ and $t_0/T_s = 1 - \left(\frac{3}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta\right)/V_{\mathrm{dc}}$, Table 4 gives

$$\begin{aligned} d_a &= \frac{t_1 + t_2}{T_s} + \frac{t_0}{2T_s} = \frac{1}{V_{\mathrm{dc}}}\left(\frac{3}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta\right) + \frac{1}{2} - \frac{1}{2V_{\mathrm{dc}}}\left(\frac{3}{2}U_\alpha + \frac{\sqrt{3}}{2}U_\beta\right) \\ &= \frac{1}{2} + \frac{1}{V_{\mathrm{dc}}}\left(\frac{3}{4}U_\alpha + \frac{\sqrt{3}}{4}U_\beta\right), \end{aligned} \tag{32}$$

which coincides with (31). The same holds for d_b, d_c and, by the 60° symmetry of the problem, for all sectors. ■

Consequence. There is no performance argument for the sector-based implementation. The min–max form is preferred in practice; the geometric SVM description remains the natural language for reasoning about vector selection, overmodulation and multilevel extensions.

6 Implementation

6.1 Classical SVM

Listing 1: Sector-based SVM.

```
void svm_classic(float ua, float ub, float vdc, float *da, float *db, float *dc)
{
    /* phase projections (inverse Clarke) */
    const float va =  ua;
    const float vb = -0.5f*ua + 0.8660254f*ub;
    const float vc = -0.5f*ua - 0.8660254f*ub;

    /* sector from the ordering of the projections */
    int N = (va > vb) + 2*(vb > vc) + 4*(vc > va);
    static const int sect_lut[8] = {0, 6, 2, 1, 4, 5, 3, 0};
    int k = sect_lut[N];                    /* 1..6 */

    /* auxiliary variables, normalised to Ts */
    const float X =  1.7320508f*ub / vdc;
    const float Y = ( 1.5f*ua + 0.8660254f*ub) / vdc;
    const float Z = (-1.5f*ua + 0.8660254f*ub) / vdc;

    float t1, t2;
    switch (k) {
        case 1: t1 = -Z; t2 =  X; break;
        case 2: t1 =  Y; t2 =  Z; break;
```

```
        case 3: t1 =  X; t2 = -Y; break;
        case 4: t1 =  Z; t2 = -X; break;
        case 5: t1 = -Y; t2 = -Z; break;
        default:t1 = -X; t2 =  Y; break;
    }

    /* overmodulation: scale back onto the hexagon boundary */
    float sum = t1 + t2;
    if (sum > 1.0f) { t1 /= sum; t2 /= sum; sum = 1.0f; }

    const float t0h = 0.5f*(1.0f - sum);    /* t0/(2*Ts) */

    switch (k) {
        case 1: *da = t1+t2+t0h; *db = t2+t0h;    *dc = t0h;       break;
        case 2: *da = t1+t0h;    *db = t1+t2+t0h; *dc = t0h;       break;
        case 3: *da = t0h;       *db = t1+t2+t0h; *dc = t2+t0h;    break;
        case 4: *da = t0h;       *db = t1+t0h;    *dc = t1+t2+t0h; break;
        case 5: *da = t2+t0h;    *db = t0h;       *dc = t1+t2+t0h; break;
        default:*da = t1+t2+t0h; *db = t0h;       *dc = t1+t0h;    break;
    }
}
```

Note that in the second switch the on-time associated with the *first* vector of the sector, t_1, and that of the *second*, t_2, swap roles between odd and even sectors, consistently with Table 4.

6.2 Min–max injection

Listing 2: Carrier-based SVM with triangular injection.

```
void svm_minmax(float ua, float ub, float vdc, float *da, float *db, float *dc)
{
    /* inverse Clarke */
    float va =  ua;
    float vb = -0.5f*ua + 0.8660254f*ub;
    float vc = -0.5f*ua - 0.8660254f*ub;

    /* min and max of the three references */
    float vmax = va > vb ? (va > vc ? va : vc) : (vb > vc ? vb : vc);
    float vmin = va < vb ? (va < vc ? va : vc) : (vb < vc ? vb : vc);

    /* triangular zero sequence */
    float v0 = -0.5f*(vmax + vmin);

    const float inv = 1.0f / vdc;
    *da = 0.5f + (va + v0)*inv;
    *db = 0.5f + (vb + v0)*inv;
    *dc = 0.5f + (vc + v0)*inv;

    /* saturation (enters overmodulation) */
    *da = *da < 0.0f ? 0.0f : (*da > 1.0f ? 1.0f : *da);
    *db = *db < 0.0f ? 0.0f : (*db > 1.0f ? 1.0f : *db);
    *dc = *dc < 0.0f ? 0.0f : (*dc > 1.0f ? 1.0f : *dc);
}
```

7 Practical notes

- **Timer configuration.** Use a centre-aligned (up–down) counter with period N and load $\mathrm{CCR}_x = (1 - d_x)\,N$ or $d_x N$ depending on the output polarity. Update the compare registers through the preload registers, synchronised with the counter underflow, to avoid glitches.

- **Sampling.** Sample the currents at the counter underflow (and, for double-update schemes, also at overflow): at those instants a zero vector is applied and the sampled value is the average of the ripple.
- **Dead time.** Dead time introduces a volt-second error whose sign depends on the current polarity, producing 5th and 7th harmonics and distortion around the current zero crossings. Compensate either in the duty cycles or in the voltage reference.
- **Minimum pulse width.** Pulses shorter than the device transition time plus the dead time cannot be reproduced. Either suppress them (pulse dropping) or merge them, and account for the resulting voltage error.
- **DC-link feedforward.** Use the measured V_{dc} in (25) so that DC-link ripple does not propagate into the output voltage.
- **Overmodulation.** Beyond $|\boldsymbol{u}^*| = V_{\text{dc}}/\sqrt{3}$ the reference leaves the inscribed circle. Simple saturation of the duty cycles gives a smooth but nonlinear transfer characteristic; dedicated mode-I / mode-II algorithms preserve the fundamental gain up to six-step operation ($\hat{U} = \frac{2}{\pi}V_{\text{dc}}$), at the cost of low-order harmonics. Anti-windup on the current controllers is mandatory in this region.
- **Discontinuous PWM.** Replacing (24) with $u_0 = \frac{V_{\text{dc}}}{2}\,\text{sgn}(u_{\max}) - u_{\max}$ (or the analogous expression with $u_{\min}$) clamps one leg at a time, reducing switching losses. The clamping window can be aligned with the current peak to maximise the benefit, which is useful at high modulation index.

8 Summary

	Classical SVM	**Min–max injection**
Sector detection	required	not required
Trigonometry	optional (sign method avoids it)	none
Operations per period	∼20 plus branches	∼12, branch-light
Zero-vector split	explicit (k_0)	implicit ($k_0 = 0.5$)
Linear limit $\hat{U}_{\text{ph}}$	$V_{\text{dc}}/\sqrt{3}$	$V_{\text{dc}}/\sqrt{3}$
Output waveform	identical for $k_0 = 0.5$	
Best suited to	vector reasoning, DPWM variants, multilevel	production firmware

Table 5: Comparison of the two implementations.

Both methods start from the same pair (U_α, U_β) and end with three duty cycles in $[0, 1]$. The classical formulation makes the geometry explicit and is the right framework for overmodulation and multilevel topologies; the min–max formulation is the compact and numerically robust way of implementing it.